TryHackMe Writeup: Networking Concepts

Field	Details
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Intern ID	DG/15/OCT/CYBER/040
Program	DG Interns Hub, Week 1
Room	Networking Concepts
Room link	https://tryhackme.com/room/networkingconcepts?utm_campaign=social_share&utm_medium=social&utm_content=share-completed-room&utm_source=copy&sharerId=698d81bc7977c1741b32e1c8
Estimated time	60 min
Status	Completed (100%)

Objective

Learn about the ISO OSI model and the TCP/IP protocol suite in more detail.

What I did, task by task

ISO OSI model

- Went through each of the seven layers and what it is responsible for, with the protocols and devices that work at each layer.

TCP/IP model

- A simpler four layer model (Application, Transport, Internet, Network Access) that the real internet is built on.
- Compared how the layers of OSI map to the layers of TCP/IP.

Encapsulation and decapsulation

- Each layer adds its own header to the data on the sender side, and each layer removes it on the receiver side.

Protocols and transport

- Saw how common protocols like HTTP, DNS, FTP and SMTP sit on top of TCP or UDP.
- Revised the TCP three way handshake and how ports identify services.

Key takeaways

- OSI is the reference model for understanding, TCP/IP is what is actually used.
- Knowing which layer a protocol or attack belongs to helps in analysing alerts.
- Headers carry the information each layer needs, such as addresses and ports.

Why this matters for a SOC role

Many detections are written around specific layers, like port scans at the transport layer or ARP spoofing at the data link layer. This room made that mapping clearer.

Proof of completion

